

22CH203 & Engineering Chemistry II

1. * The reaction with lower activation energy (low peak) is faster because the reaction occurs more easily.
* The reaction with larger energy difference is more stable because the products have lower energy.
* Thus, low activation energy increases speed, while large energy difference increases stability
2. * The reaction releases a large amount of energy rapidly, so it is hazardous and hard to control
* $\therefore$ It needs strict safety precautions and controlled conditions
3. * All substances are in same liquid, so reaction happens faster
* Catalyst mixes completely and increases reaction efficiency
* Intermediate substance forms temporarily, showing the reaction occurs step by step before final product form
4. * After a certain limit, adding more catalyst does not increase the reaction rate because reactant concentration becomes the limiting factor
* The system reaches saturation, so the reaction rate becomes almost constant
5. * Low activation energy process starts easily and is safer but gives less energy output
* High activation energy process needs more initial energy, but produces higher energy and better productivity
* For large-scale continuous production, the high energy output process is preferred if safety and cost are properly managed

6. * The reaction is an exothermic reaction because the products have lower energy than the reactants.
* The single peak ~~character~~ represents the activation energy needed for the reaction to occur.
7. * Solvent polarity changes the reaction speed by affecting reactant-catalyst interaction.
* Higher catalyst concentration increases reaction rate.
* Solvent and catalyst together decide which product is formed more.
8. * The homogeneous catalytic system gives high reaction efficiency and better product yield.
* However, catalyst recovery is difficult and increases process cost and complexity.
9. * The forward reaction has lower activation energy, so it occurs faster and more easily.
* The reverse reaction has higher activation energy, so it proceeds more slowly.
* Because of this difference, the reaction initially favours product formation.
* At equilibrium, both forward and reverse reactions occur at equal rates, but concentrations remain constant.
* Lower energy products are more stable, making the reaction more feasible and thermodynamically favourable.
10. * The powdered catalyst has a large surface area compared to large solid chunks.
* Most surface area provides more active sites for reactant molecules to absorb and react.

* This increases the frequency of effective collisions and speed up the reaction.

* In large chunks, only a small outer surface is available, so the reaction rate is slower.

* $\therefore$ catalytic efficiency increases with increase in surface area and availability of active sites.

11. * The energy barrier must be crossed before reactants form products.

* Even though the reaction is thermodynamically favorable, high activation energy makes the reaction slow at room temperature.

* The peak in the energy profile represents the high-energy intermediate state.

* Increasing temperature gives more energy to cross the barrier.

* $\therefore$ the no. of effective collisions increases and the reaction rate becomes faster at high temperature.

12. * In heterogeneous catalysis, reactants first adsorb on the catalytic surface, then react and finally products are formed.

* Increased surface roughness gives more active sites, so adsorption and reaction rate increases.

* Moderate temperature increases collision frequency and reaction rate.

* Excess temperature causes side reactions and catalyst deactivation, reducing efficiency.

* The surface reaction at active sites controls the overall reaction.

13. * The enzyme is substrate-specific, and forms only the required product.
* Small amounts of enzyme give a fast reaction showing high efficiency.
* The enzyme is not consumed and can be reused repeatedly.
* It increases reaction rate but does not change equilibrium position.
* High selectivity, reusability, and efficiency make enzymes useful in industries.
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14. * Substance increasing the rate is catalyst, the one decreasing the rate is an inhibitor.
* Catalysts lower activation energy and speed up the reaction.
* Inhibitor slows the reaction by interfering with the mechanism.
* They change only the reaction rate, not the equilibrium composition.
* Impurities affect catalyst performance, so careful control is needed.
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15. * The iron catalyst provides an alternative reaction pathway with lower activation energy.
* Lower activation energy increases effective collisions and speeds up the reaction.
* This catalyst increases reaction speed without changing temperature or pressure.
* The catalyst remains chemically unchanged after the reaction and can be reused.
* The lower peak in energy profile shows reduced activation energy in the catalysed reaction.

16. * Homogenous catalysis (same phase) gives fast, uniform reaction but catalyst separation is difficult
- * Heterogenous catalysis (solid catalyst + gas reactants) occurs on solid surface and is easy to separate and reuse
- * Homogenous system is more efficient but less practical for large-scale production.
- * Heterogenous catalysis is slightly slower but more economical and scalable.
- * Hence, heterogenous catalysis is preferred for industrial use.
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17. * The mass difference is called mass defect.
- * It is converted into nuclear binding energy that holds nucleons together.
- * Greater mass defect means higher stability and stronger nucleus
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18. * They are isotopes, so they have the same number of electrons and electronic configuration.
- * Chemical reactions depend on electrons, not mass
- * So their chemical behaviour remains the same
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19. * The graph shows radioactive decay, where unstable nuclei break down over time
- ~~* The rapid initial drop~~
- * Initially decay is fast because many nuclei are present
- * Later it slows down but never becomes zero

20. * Mean life is the average time a nucleus exists before decaying.

* Mean life is related to half life by

$$\text{Mean life} = \frac{\text{Half life}}{0.693} \quad (\text{or}) \quad \tau = \frac{T_{\frac{1}{2}}}{0.693}$$

21. * Both isotopes have same electrons, so their chemical behaviour is the same

* They differ in nucleus, so their stability and radius emission are different.

* The radioactive isotope emits detectable radiation, so it is used in medical diagnosis, while the stable one is inactive.

22. * Too many neutrons create imbalance between nuclear forces and repulsion between protons.

* This weakens nuclear binding, making the nucleus unstable and more likely to decay.

23. * Radioactive decay is a nuclear process, so it does not depend on temperature or pressure

* External conditions affect electrons, not the nucleus where decay occurs

* Hence, decay rate constant, ensuring reliable use in medical tracers

24. * One-fourth means the sample has undergone

two half-lives ($\frac{1}{2} \rightarrow \frac{1}{4}$)

* So, 2 decay stages are involved

25. * Chlorine isotopes have same electrons, so they show same chemical behaviour and reaction pathway
- * They differ in mass due to different neutrons.
 - * Mass difference causes a slight change in reaction rate which is called isotope effect
 - * ~~Physical~~ Physical properties like diffusion and speed are affected but not chemical behaviour
 - * Hence, Isotopes behave similarly chemically but differ slightly in physical behaviour
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26. * The reaction follows conservation of mass number and atomic number
- * A new element forms due to change in proton number after bombardment
 - * Energy change is calculated using mass defect
 - * The mass defect is converted to energy using $E = mc^2$
 - * Hence, Mass loss directly converts into released nuclear energy
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27. * Based on Graham's law, lighter isotopes diffuse faster than heavier ones
- * The difference in mass cause different diffusion rates
 - * At constant conditions, diffusion depends only on molar mass

- * Faster diffusion of lighter isotope allows separation through membranes.
- * Repeated diffusion steps help in efficient industrial isotope separation.

28. * The reaction is nuclear fusion, where light nuclei combine to form a heavier nuclei.

- * very high temperature and pressure are needed to overcome electrostatic repulsion between nuclei.

- * The process releases large energy due to mass defect converted into energy ($E=mc^2$).

- * Energy output is much heavier than chemical reactions, making it highly efficient.

- * However, maintaining extreme conditions continuously is difficult, so large-scale sustained use is challenging.

29. * At high temperature, intrinsic carrier concentration increases in the narrow band gap semiconductor.

- * More electrons jump from valence band to conduction band, increasing conductivity and leakage current.

- * This causes loss of proper switch behavior.

- * Temperature changes the carrier distribution, increasing thermally generated carriers.

- * The Fermi level shifts toward the ~~intra~~ intrinsic level, reducing control over conductivity.

30. * For an n-type semiconductor, electrons are the majority carriers, so drift conduction is mainly due to electrons

* Electrical conductivity (σ) = nqu

* Given: $n = 10^{16} \text{ cm}^{-3}$, $\mu = 1400 \text{ cm}^2/\text{Vs}$

We know that: $q = 1.6 \times 10^{-19} \text{ C}$

$$\sigma = nqu = 10^{16} \times 1.6 \times 10^{-19} \times 1400$$

$$\sigma = 10^{16} \times 1.6 \times 10^{-19} \times 14 \times 10^2$$

$$\sigma = 1.6 \times 14 \times 10^{-19+2+16} = 1.6 \times 14 \times 10^{-1} = \frac{1.6 \times 14}{10}$$

$$\boxed{\sigma = 2.24 \text{ S/cm}}$$
 It is the conductivity of the semiconductor

* Donor atoms release free electrons, so electrons carry most of the current
* very few holes are present, so their effect on conduction is very small.

31. * The wire should use a conductor because it has free electrons which allow easy current flow

* The switching element should use a semiconductor because conductivity can be controlled

* The insulating cover should use an insulator because its band gap prevents electron flow

* conductors have high conductivity, semiconductors have moderate and controllable conductivity and insulators have very low conductivity

* Hence, metals are used for wires, semiconductors for switches and rubber/plastic for insulation

32. * The semiconductor is p-type due to acceptor impurities, where holes are the majority carriers

* At low temperature, thermal energy is insufficient to fully ionize acceptor atoms.

* Fewer free holes are produced, so conductivity decreases and resistivity increases.

* carrier scattering is low, but carrier generation is strongly limited

* Hence, incomplete ionization of impurities is the dominant factor reducing carrier transport at low temperature
